Lab 5
Basic Logic Gate Implementation
Part 5

Prof. Chih-Cheng Tseng (曾志成)

cctseng@mail.ntou.edu.tw

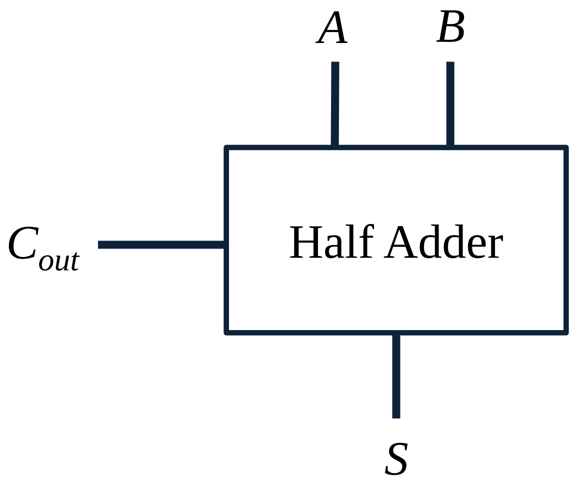
Contents of this slide are provided by NTOU VLSI Lab601

Objectives

- Learn the operation principle of adders.

1-bit Half Adder

- The half adder only considers the two numbers being added and does not account for carry from the previous stage.



S

$A \backslash B$	0	1
0	0	1
1	1	0

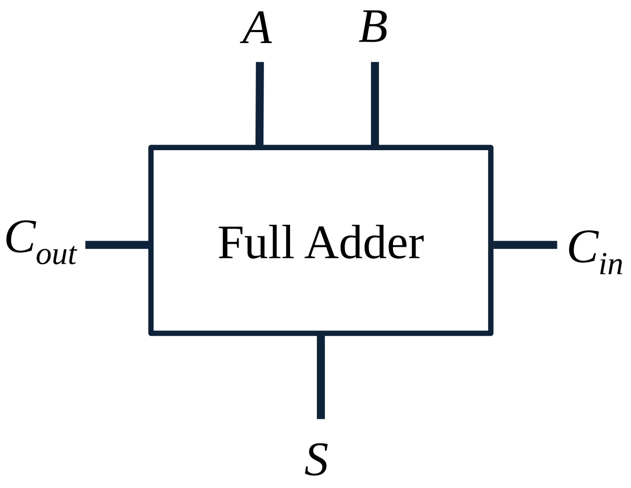
C_{out}

$A \backslash B$	0	1
0	0	0
1	0	1

Truth Table			
Input		Output	
A	B	C_{out}	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

1-bit Full Adder

- The full adder accounts for carry from the previous stage.



S

A \ BC_{in}	00	01	11	10
0	0	1	0	1
1	1	0	1	0

C_{out}

A \ BC_{in}	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Truth Table				
Input			Output	
A	B	C_{in}	C_{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

LAB 5-1

- Implement the half adder circuit.
  1. Draw the truth table.
  2. Draw the Karnaugh map (K-map).
  3. Simplify the Boolean algebra to its simplest form.
  4. Draw the logic circuit diagram.
  5. Implement this logic circuit on a breadboard.
  6. Take pictures of each experiment result.

- Please add LEDs to the output.

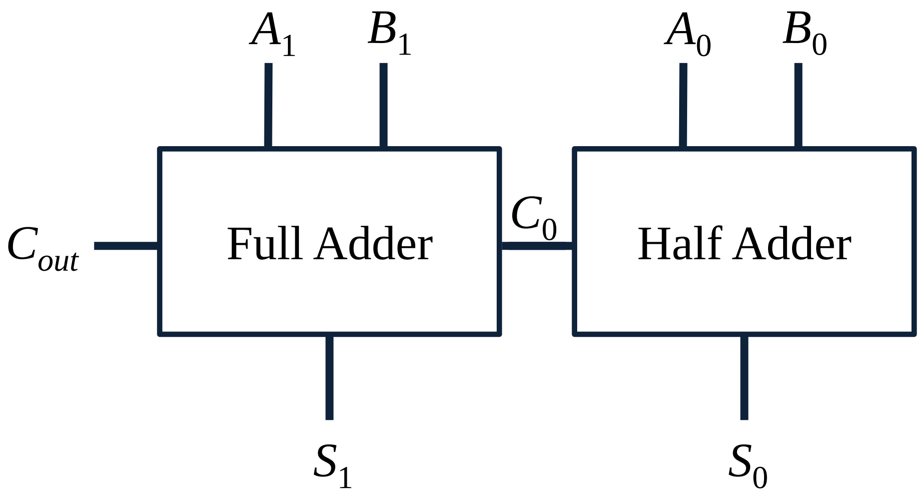
LAB 5-2

- Implement the full adder circuit.
  1. Draw the truth table.
  2. Draw the Karnaugh map (K-map).
  3. Simplify the Boolean algebra to its simplest form.
  4. Draw the logic circuit diagram.
  5. Implement this logic circuit on a breadboard.
  6. Take pictures of each experiment result.

- Please add LEDs to the output.

LAB 5-3

- Using 1-bit half adder and 1-bit full adder to design a 2-bit adder.
  1. Draw the truth table.
  2. Draw the Karnaugh map (K-map).
  3. Simplify the Boolean algebra to its simplest form.
  4. Draw the logic circuit diagram.
  5. Implement this logic circuit on a breadboard.
  6. Take pictures of each experiment result.



- Please add LEDs to the output.

Bonus

- Design a 2-bit subtractor by using the 1-bit full adder. (Tips: $A - B = A + $ (2's complement of B).)
  1. Draw the truth table.
  2. Simplify the Boolean algebra to its simplest form.
  3. Draw the logic circuit diagram.
  4. Implement this logic circuit on a breadboard.

- Please add LEDs to the output.